

Week 4 Example

```
library(tidyverse) # AI-EDIT(2026-06-23): tidyverse-default – consolidated core
library() calls into library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1      ✓ readr      2.2.0
✓ forcats    1.0.1      ✓ stringr    1.6.0
✓ ggplot2    4.0.3      ✓ tibble     3.3.1
✓ lubridate  1.9.5      ✓ tidyr      1.3.2
✓ purrr      1.2.2

— Conflicts — tidyverse_conflicts()
—
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
candidate_paths <- c(
  "examples/nhanes-equity/data/nhanes_equity_v6.csv",
  "../examples/nhanes-equity/data/nhanes_equity_v6.csv"
)

nhanes_path <- candidate_paths[file.exists(candidate_paths)][1]
if (is.na(nhanes_path)) {
  stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
}

nhanes_analysis <- read_csv(nhanes_path, show_col_types = FALSE)

nhanes_analysis |>
  summarise(
    rows = n(),
    missing_bmi = sum(is.na(BMI)),
    missing_income = sum(is.na(IncomeGroup))
  )
```

```
# A tibble: 1 × 3
  rows missing_bmi missing_income
```

	<int>	<int>	<int>
1	105626	9356	9415

```
nhanes_analysis |>
  filter(!is.na(BMI), !is.na(IncomeGroup), Age >= 20, Age <= 80) |>
  group_by(IncomeGroup) |>
  summarise(
    n = n(),
    mean_bmi = round(mean(BMI), 1),
    .groups = "drop"
  )
```

```
# A tibble: 3 × 3
  IncomeGroup      n mean_bmi
  <chr>      <int>   <dbl>
1 High Income (>3.5) 16330    28.6
2 Low Income (<1.3) 15293    29.4
3 Middle Income    19228    29.3
```